

# القوانين

Sec (4.4) :

$$1] \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \quad \text{g or } \frac{0}{0}$$

$$2] \lim_{x \rightarrow a} f(x) g(x) = \lim_{x \rightarrow a} f(x) \text{ or } g(x) \quad 0, \pm \infty$$

$$\lim_{x \rightarrow a} \frac{f(x)}{\frac{1}{g(x)}} \leftarrow f(x)$$

$$3] \lim_{x \rightarrow a} [f(x) \pm g(x)] =$$

$$\lim_{x \rightarrow a} f(x) [1 + \frac{g(x)}{f(x)}] \quad \text{على شكل } \frac{0}{0} \text{ } (\infty - \infty)$$

$$4] \lim_{x \rightarrow a} f(x)^{g(x)} =$$

$$y = f(x)^{g(x)}$$

$$\ln y = g(x) \ln f(x)$$

$$\lim_{x \rightarrow a} \ln y = \lim_{x \rightarrow a} g(x) \ln f(x)$$

$$= \lim_{x \rightarrow a} \frac{\ln f(x)}{\frac{1}{g(x)}} = m$$

$$\therefore \lim_{x \rightarrow a} f(x)^{g(x)} = e^m$$

Sec (7.1) :-

$$\int u dv = uv - \int v du$$

Sec (7.2) :-

$$\textcircled{1} \int \cos^a x \sin^b x dx$$

$$\Rightarrow \sin^2 x = 1 - \cos^2 x$$

$$u = \cos x \quad du = -\sin x$$

$$\Rightarrow \cos^2 x = 1 - \sin^2 x$$

$$u = \sin x \quad du = \cos x$$

$$\Rightarrow \sin(2x) = 2 \sin x \cos x$$

$$\Rightarrow \cos^2 x = \frac{1}{2} (1 + \cos(2x))$$

$$\Rightarrow \sin^2 x = \frac{1}{2} (1 - \cos(2x))$$

$$\Rightarrow \int \sin(ax) = -\frac{\cos(ax)}{a} + C$$

$$\Rightarrow \int \cos(ax) = \frac{\sin(ax)}{a} + C$$

$$\Rightarrow \sin^2 x + \cos^2 x = 1$$



$$\textcircled{2} \int \tan^a x \sec^b x dx$$

$$\Rightarrow \sec^2 x = 1 + \tan^2 x$$

$$u = \tan x \quad du = \sec^2 x$$

$$\Rightarrow \tan^2 x = \sec^2 x - 1$$

$$u = \sec x \quad du = \sec x \tan x$$

$$\textcircled{3} \int \cot^a x \csc^b x dx$$

$$\Rightarrow \csc^2 x = 1 + \cot^2 x$$

$$u = \cot x \quad du = -\csc^2 x$$

$$\Rightarrow \cot^2 x = \csc^2 x - 1$$

$$u = \csc x \quad du = -\csc x \cot x$$

$$\textcircled{4} \int \cos(mx) \sin(nx) dx$$

$$\sin(mx) \cos(nx) = \frac{1}{2} [\sin(m-n)x + \sin(m+n)x]$$

$$\textcircled{5} \int \cos(mx) \cos(nx) dx$$

$$\cos(mx) \cos(nx) = \frac{1}{2} [\cos(m-n)x + \cos(m+n)x]$$

$$\textcircled{6} \int \sin(mx) \sin(nx) dx$$

$$\sin(mx) \sin(nx) = \frac{1}{2} [\cos(m-n)x - \cos(m+n)x]$$

## Sec (7.3) :

① اِستبدال  $x \leftarrow \theta$

$$\sqrt{x^2 - a^2}$$

$$x = a \sec \theta$$

$$\sqrt{x^2 + a^2}$$

$$x = a \tan \theta$$

$$\sqrt{a^2 - x^2}$$

$$x = a \sin \theta$$

⇒ Complete square

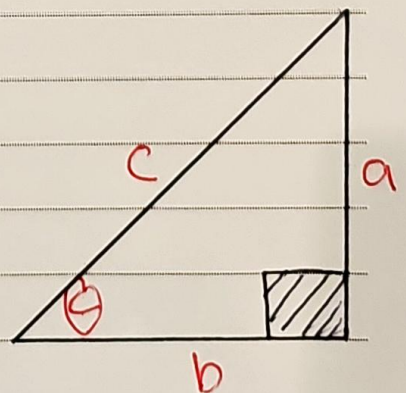
$$① x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

$$② ax^2 + bx + c = a \left[ x^2 + \frac{bx}{a} + \frac{c}{a} \right]$$

$$\Rightarrow \sin \theta = \frac{a}{c} \Rightarrow \csc \theta = \frac{c}{a}$$

$$\cos \theta = \frac{b}{c} \Rightarrow \sec \theta = \frac{c}{b}$$

$$\tan \theta = \frac{a}{b} \Rightarrow \cot \theta = \frac{b}{a}$$





$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$\int \frac{x}{\sqrt{a^2 - x^2}} dx = -\sqrt{a^2 - x^2} + C$$

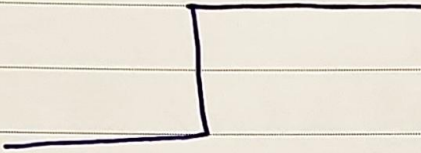
$$(2) \int \frac{x}{\sqrt{x^2 \pm a^2}} dx = \sqrt{x^2 \pm a^2} + C$$

$$\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln |x + \sqrt{x^2 \pm a^2}| + C$$

## Sec (7.4) %

اذا كان عددي  $\frac{A}{B}$  or  $\frac{C}{D}$  حدود  $\frac{A}{B}$   $\frac{C}{D}$

[1] مقام  $\geq$  بس



$$\int \frac{dx}{\text{مقام}} = \int \text{تابع} dx + \int \frac{3L}{\text{مقام}} dx$$

[2] مقام  $<$  بس



$$\Delta > 0$$

p.f

$$\frac{A}{(x-a)(x-b)} = \frac{A}{(x-a)} + \frac{B}{(x-b)}$$

$$(x-a)(x-b) = A(x-b) + B(x-a)$$

$$a+x = A$$

$$a+x = B$$

$$\Delta < 0$$

Compl. sq

$$\left(x + \frac{b}{2}\right)^2 + \left(\frac{b}{2}\right)^2 + C$$

$$\Delta = 0$$

repeated

$$a(x-x_1)^2$$

$$u = x - 1$$



[3] W. S.

$$\int \frac{\alpha \sin x + \beta \cos x + C}{a \sin x + b \cos x + C} dx$$

$$\textcircled{1} u = \tan\left(\frac{x}{2}\right)$$

$$\textcircled{2} \sin x = \frac{2u}{1+u^2}, \quad \cos x = \frac{1-u^2}{1+u^2}$$

$$= \frac{2}{1+u^2}$$

[4]  $\int \frac{\text{حدودية}}{\pm \sqrt{\quad}} dx$

$$\textcircled{1} \int \frac{\text{حدودية}}{\pm \sqrt{\quad}} dx$$

$$\textcircled{2} \text{ L.C.D.}$$

$$\begin{array}{cc} 6 & 2 \\ \wedge & \wedge \\ 2 & 3 \end{array}$$

اخذ صغرى الاعداد والاعلى

$$\text{LCD} = 2 \cdot 3 = 6$$

$$\textcircled{3} u^m = \left( \frac{1}{2} \right)^{\frac{1}{2}}$$

## Sec (7.8):

$$\boxed{\text{I}} \int_{-\infty}^a f(x) dx \text{ or } \int_a^{\infty} f(x) dx \text{ or } \int_{-\infty}^{\infty} f(x) dx$$

$$\Rightarrow \int_{-\infty}^a f(x) dx = \lim_{t \rightarrow -\infty} \int_t^a f(x) dx \quad \left. \begin{array}{l} \# \rightarrow \boxed{C} \\ \rightarrow \end{array} \right\}$$

$$\Rightarrow \int_a^{\infty} f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx \quad \left. \begin{array}{l} \rightarrow \infty \rightarrow \boxed{D} \end{array} \right\}$$

$$\Rightarrow \int_{-\infty}^{\infty} f(x) dx = \underbrace{\int_{-\infty}^a f(x) dx}_{\text{"1"}} + \underbrace{\int_a^{\infty} f(x) dx}_{\text{"2"}}$$

$$C + C = C$$

$$D + D = D$$

$$D + C = D$$



## Sec (11.1) :

1] using  $\lim$  of function :

$$x \leftarrow n \text{ ju! } ①$$

$$② \lim_{x \rightarrow \infty} f(x) =$$

$$③ \therefore \lim_{x \rightarrow \infty} f(x) = * \Rightarrow \lim_{n \rightarrow \infty} a_n = *$$

2] Squeeze Theorem :

$$① \leq b_n \leq$$

$$\downarrow$$
$$\lim_{n \rightarrow \infty} = *$$

$$\downarrow$$
$$\lim_{n \rightarrow \infty} = *$$

$$\therefore \lim_{n \rightarrow \infty} b_n = *$$

3] using  $\lim_{n \rightarrow \infty} |a_n| = 0$  :

$$\lim_{n \rightarrow \infty} |a_n| = 0 \Rightarrow \lim_{n \rightarrow \infty} a_n = 0$$

4]  $a_n = (-1)^n b_n$  :

$$\lim_{n \rightarrow \infty} b_n = 0$$

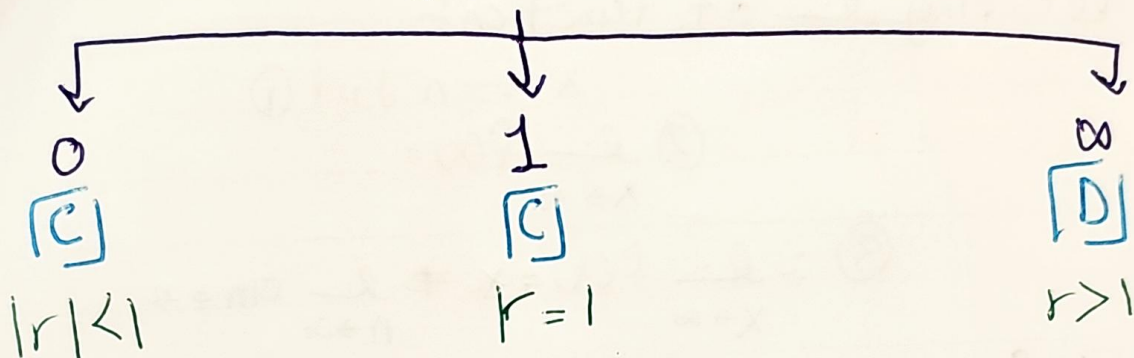
[C]

$$\lim_{n \rightarrow \infty} b_n = \infty$$

[D]

$$a_n = r^n \cdot \sigma$$

$$\textcircled{1} \lim_{n \rightarrow \infty} r^n =$$



[6] recurrence relation

$$\lim_{n \rightarrow \infty} a_{n+1} = \lim_{n \rightarrow \infty} a_n$$

$$\Rightarrow \sum_{n=0}^{\infty} \underbrace{\left(\frac{2}{3}\right)^n}_K \underbrace{\left(\frac{-1}{2}\right)^n}_u$$

$$S = (1+4)^K$$



## Sec (11.2) :

### [1] Telescoping S. :

$$\sum (x_j - x_{j-1})$$

①  $a_n =$

②  $S_n = a_1 + a_2 + a_3 + \dots + a_{n-1} + a_n$

③  $S_n$  محصية

④  $\lim_{n \rightarrow \infty} S_n =$

$\rightarrow \# \quad \therefore \sum a_n = C$

$\rightarrow \pm \infty \quad \therefore \sum a_n = D$

### [2] Geometric S. :

$$\sum_{n=1}^{\infty} a r^{n-1}$$

$|r| < 1$  [C]

$$S = \frac{a}{1-r}$$

$|r| > 1$  [D]

No Sum.

## Sec (11.3 + 11.4) %

### 1] I. L %

اذا وجد تكامل

①  $f$  is  $\in [1, \infty)$

②  $f$  is  $\boxed{C} \in [1, \infty)$

③  $f$  is  $\downarrow \in [1, \infty)$

$\hookrightarrow f'(x) < 1$

$$\therefore \int_1^{\infty} f(x) dx = \sum_{n=1}^{\infty} a_n$$

### 2] P-Series %

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

$p > 1$

C

$p \leq 1$

D

### 3] B. C. T %

①  $\ln(n) \ll n^a \ll a^n \ll n! \ll n^n$

②  $0 < \sin^2(n) \leq 1$  }  $\cos, \cos^2$  في  
 $-1 < \sin(n) \leq 1$

D  $\rightarrow$  D

③  $\pi/4 \leq \tan^{-1}(n) < \pi/2$

$0 \leq \tanh(n) \leq 1$

C  $\leftarrow$  C



L.C.T. 2

$$\sum \frac{\text{دالة or } \sin(\cdot)}{\text{دالة or } \sin(\cdot)}$$

①  $b_n =$  استجابة

②  $\sum b_n = \sum$  by p.s or G.S. ولاحقاً

③  $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = a_n \cdot \frac{\text{دالة}}{b_n} > 0$

$\therefore \sum a_n$  is by L.C.T.

Sec (11.5 + 11.6) %

① A.C. + %

$$\Rightarrow \sum_{n=1}^{\infty} (-1)^n b_n$$

①  $\lim_{n \rightarrow \infty} b_n = 0$

$\therefore \lim_{n \rightarrow \infty} b_n = \infty$  D

②  $[b_n] \downarrow$

على الوجود

$f(x) < 0$   
 $\frac{1}{\infty}$

$b_{n+1} \leq b_n$  (n!)  
 $0 < b_n <$

$\therefore \sum a_n$  is by A.C. +

② A.C, C.C, Div %

$\sum |a_n| = \text{C}$  C  $\xrightarrow{\text{by L.C. + or B.C. + P or G.S}}$   $\sum |a_n| = P$   
 $\therefore \text{A.C}$

①  $\lim_{n \rightarrow \infty} |a_n| = 0$

①  $\lim_{n \rightarrow \infty} |a_n| \neq 0$

②  $b_n \downarrow$   
C.C

div



ition 0

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \text{خاص } n!, n^p, p^n, n^n$$

①  $a_n =$

$a_{n+1} =$

②  $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = a_{n+1} \cdot \frac{1}{a_n}$  مقلوب

③ افكك  $n$  و  $1$  حسب

$< 1$   
C

$> 1$  or  $\infty$   
D

$= 1$   
X

[4] root. T 0

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} =$$

①  $a_n = ( )^n$

②  $\lim_{n \rightarrow \infty} \sqrt[n]{( )^n} = \lim_{n \rightarrow \infty} a_n$  p.s or G.S or ...

$< 1$   
C

$> 1$  or  $\infty$   
D

$= 1$   
X

## Sec (6.2) &

### [1] Disc &

$$V = \pi r^2 \cdot \text{Thic} \begin{matrix} \nearrow dx \\ \searrow dy \end{matrix}$$

$$V = \pi \int r^2 \cdot \begin{matrix} dx \\ \text{or} \\ dy \end{matrix}$$

### [2] Washer &

$$V = \pi \int [R^2 - r^2] \cdot \text{Thic} \begin{matrix} \nearrow dx \\ \searrow dy \end{matrix}$$

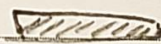
$R^2$  = بعد - محور عارضة أو

تحت - فوق

$r$  = قريب - عارضة محور or  
تحت - فوق



$dx$



$dy$

### [3] Shell &

$$V = 2\pi r h \cdot \text{Thic} \begin{matrix} \nearrow dx \\ \searrow dy \end{matrix}$$

$\nwarrow$   $\nearrow$   
 ر - يار - محيط      تحت - فوق



① نرم

② نرم عبور در راه

③ مختار عاصور "لازم يكون عاصوري على عبور دوا" ←

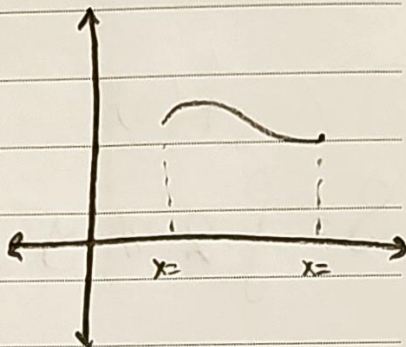
Shell صوازي لعبور الدوله

④ نقاط التقاطع

⇒ Sec (8.1 + 8.2) %

①  $y = f(x)$  on  $[a, b]$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$



②  $x = f(y)$  on  $[a, b]$

$$L = \int_a^b \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

⇒ Rotation y-axis %

①  $y = f(x)$

$$S = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

②  $x = f(y)$

$$S = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$



⇒ Rotation x-axis o

①  $y = f(x)$

$$S = \int 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

②  $x = f(y)$

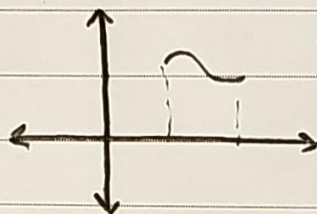
$$S = \int 2\pi y \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

## Sec (8.3) %

⇒ Centroid  $(\bar{x}, \bar{y})$  %

$$\textcircled{1} A = \int_a^b f(x) dx$$

$$\textcircled{2} \bar{x} = \frac{1}{A} \int_a^b x f(x) dx$$



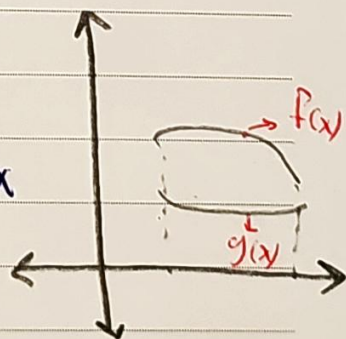
$$\textcircled{3} \bar{y} = \frac{1}{2A} \int_a^b f^2(x) dx$$

⇒ Centroid  $(\bar{x}, \bar{y})$  %  $\therefore P(\bar{x}, \bar{y}) =$

$$\textcircled{1} A = \int_a^b [f(x) - g(x)] dx$$

$$\textcircled{2} \bar{x} = \frac{1}{A} \int_a^b x [f(x) - g(x)] dx$$

$$\textcircled{3} \bar{y} = \frac{1}{2A} \int_a^b [f(x)^2 - g(x)^2] dx$$





## Sec (10.1 + 10.2) :-

$$\textcircled{1} \cos^2 t + \sin^2 t = 1$$

$$\textcircled{2} (x-a)^2 + (y-b)^2 = c$$

center =  $(a, b)$       radius =  $\sqrt{c}$

### [1] Areas :-

$$\textcircled{1} A = \int_a^b x(t) y'(t) dt$$

$$\textcircled{2} A = \int_a^b y(t) x'(t) dt$$

### [2] Arc length :-

$$L = \int_a^b \sqrt{(x')^2 + (y')^2} dt$$

### [3] Tangent lines :-

$$y - y_{(t_0)} = m (x - x_{t_0})$$

Surface area of

① rotation x-axis

$$A = 2\pi \int y \sqrt{(\dot{x})^2 + (\dot{y})^2} dt$$

② rotation y-axis

$$A = 2\pi \int x \sqrt{(\dot{x})^2 + (\dot{y})^2} dt$$



→ انتقالات  $(r, \theta)$  ، تطبيقات

Sum  
①

$$y' = \frac{dy/d\theta}{dx/d\theta} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$y'' = \frac{(dy/d\theta)'}{dx/d\theta} = \left(\frac{dy}{d\theta}\right)' \cdot \frac{d\theta}{dx}$$

Find equation

1] tangent line

$$y - y_1 = m(x - x_1)$$

2] normal

$$y - y_1 = \frac{1}{m}(x - x_1)$$

⇒ Can Curve up  
 $y'' > 0$

⇒ Can Curve down  
 $y'' < 0$

Find point

① stop

$$\frac{dy}{dx} = 0$$

② tang. // line

$$m_1 = m_2$$

③ tang. ⊥ line

$$m_1 = -\frac{1}{m_2}$$

④ curve v +

$$\frac{dx}{dy} = 0$$

⑤ curve H. +

$$\frac{dy}{dx} = 0$$

## sec (10.3) :

1] if  $(r, \theta)$  is given then

$$x = r \cos \theta$$

$$y = r \sin \theta$$

2] if  $(x, y)$  is given then

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1} \left( \frac{y}{x} \right)$$



## Sec (10.4) :-

[1] Arc length :-

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

[2] Area :-

$$\textcircled{1} A = \frac{1}{2} \int_a^b r^2 d\theta$$



$$\textcircled{2} A = \frac{1}{2} \int_a^b (f(\theta))^2 - (g(\theta))^2 d\theta$$

